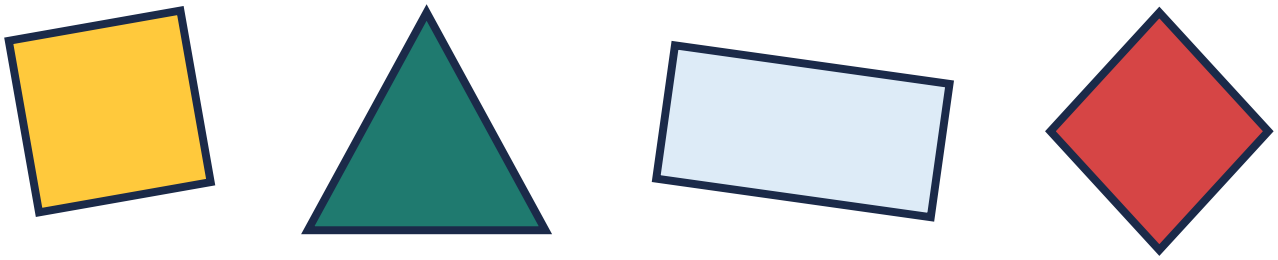
FOR PARENTS • LEVEL 2 HOLIDAY PROGRAM

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Go further questions if your child enjoys them.

The three levels

Start here is for everyone. **Think harder** stretches the idea. **Go further** and the **Investigation** are for when your child wants more, and it is fine to leave them for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Digit Detectives

LESSON 1 • NUMBERS TO 1000

Start here

1 What number is 5 hundreds, 3 tens and 8 ones?

2 In 914, what is the digit 1 worth?

3 What number is 100 more than 457?

4 What number is 10 less than 603?

Think harder

5 Use the digit cards 1, 5 and 8 once each. How many different 3-digit numbers can you make? List them in order.

6 Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?

7 What number is 3 hundreds, 14 tens and 6 ones?

What your child learns

Building 3-digit numbers from hundreds, tens and ones, finding every number from digit cards by listing in order, and spotting patterns on a 1000 chart.

Tip for you

Write three digits on paper scraps. Ask your child to make the biggest and smallest numbers, then every number, in order. Before class: cut out the digit cards (kit page) and print or draw a 1000 chart with rows of 10.

Ask

"How do you know you have found them all?"

Go further

8 Use any three of the digit cards 1, 3, 5 and 7. How many different 3-digit numbers can you make?

9 How many 3-digit numbers have digits that add up to 5?

W1 Missing Digit Mysteries

LESSON 2 • ADDITION AND SUBTRACTION WITHIN 1000

Start here

1 3A5 + 214 = 579. What is the digit A?

2 6A8 − 205 = 433. What is the digit A?

3 What number makes this true? 400 + 250 = 300 + A

4 What must be added to 365 to make 500?

Think harder

5 A magic triangle has 3 corner circles and 3 circles in the middle of the sides. Put the numbers 110, 120, 130, 140, 150 and 160 in the circles, once each, so that every side adds up to 420. Draw the triangle. Which numbers are in the corners?

6 Find the missing digits: 5A7 + B48 = 905

7 Find the missing digits: 7A1 − B58 = 263

What your child learns

Finding missing digits in sums and differences, checking by adding back, and magic squares with bigger numbers.

Tip for you

Cover one digit of a sum with your thumb. Let your child work out what is hiding, then check by adding. Before class: have paper scraps and a pencil ready for the number tiles 110 to 160.

Ask

"How can you check your answer?"

Go further

8 Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as BIG as possible.

9 The same number A goes in both places: A + 150 = 550 – A. What is A?

W2 LESSON 3 • MULTIPLICATION AND DIVISION
Legs, Wheels and Arrays

Start here

1 How many wheels are there on 5 bicycles?

2 There are 4 rows of chairs with 5 chairs in each row. How many chairs are there?

3 18 sweets are shared equally among 3 children. How many sweets does each child get?

4 How many legs do 7 hens have?

Think harder

5 A shop has some bicycles (2 wheels) and some tricycles (3 wheels), at least one of each. There are 17 wheels altogether. Find ALL the ways. Start with 1 tricycle, then 2 tricycles, and so on. Make a table or use counters. How do you know you have found them all?

6 There are 9 hens and goats on a farm. They have 24 legs altogether. How many hens and how many goats are there?

What your child learns

Arrays, equal groups and sharing with leftovers, using the 2, 3, 4, 5 and 10 times tables.

Tip for you

Count legs at the dinner table or wheels in the car park. Ask how many people or cars there must be. Before class: gather about 30 counters (buttons, dried beans or Lego bricks) and paper.

Ask

"What do you do with the ones left over?"

7 Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?

Go further

8 Ali says: “Every number from 4 to 30 can be made into an array with at least 2 rows and at least 2 counters in each row.” Is he right? Find ALL the numbers that cannot. Explain why they cannot.

9 A shop sells bicycles (2 wheels), tricycles (3 wheels) and toy cars (4 wheels). There are 9 vehicles and 28 wheels altogether, with at least one of each kind. Find ALL the ways.

W2 Pattern Machines

LESSON 4 • PATTERNS WITH MULTIPLICATION

Start here

1 Count in 4s. Fill in the next three numbers: 4, 8, 12, __, __, __

2 A pattern goes red, blue, red, blue ... What colour is the 10th bead?

3 Each hand has 5 fingers. How many fingers are on 6 hands?

4 A pattern goes star, circle, circle, star, circle, circle ... What is the 9th shape?

Think harder

5 Look at the shape pattern in the Start here page again (places 1 to 10). How many circles are there in the first 30 shapes? How do you know?

6 A pattern goes A, B, B, C, A, B, B, C ... What is the 25th letter?

What your child learns

Finding far-away places in a repeating pattern by counting in groups, and growing patterns with a table.

Tip for you

Make a pattern with spoons and forks. Ask what the 20th one will be before you build it. Before class: gather 9 red and blue cubes (or Lego bricks), about 25 lolly sticks and coloured pencils.

Ask

"How many things are in the part that repeats?"

7 Beads go red, yellow, yellow, red, yellow, yellow ... How many yellow beads are in the first 18 beads?

Go further

8 Light A flashes every 3 seconds. Light B flashes every 5 seconds. They flash together at the start. In the next 30 seconds, when do they flash together?

9 Houses are made of sticks in a row: 1 house uses 5 sticks, 2 houses use 9, 3 houses use 13. Ben has 40 sticks. What is the most houses he can make? How many sticks are left over?

W3 Mystery Number Machines

LESSON 5 • WORKING BACKWARDS

Start here

1 I think of a number. I multiply it by 5. I get 45. What is my number?

2 I think of a number. I divide it by 2. I get 8. What is my number?

3 I think of a number. I add 35. I get 90. What is my number?

4 I think of a number. I multiply it by 3, then add 2. I get 20. What is my number?

Think harder

5 A machine multiplies by a number, then adds a number. IN 2 gives OUT 11 and IN 5 gives OUT 20. Find what the machine does (try × 2, × 3, × 4 ... with IN 2, then test IN 5). Then: what OUT does IN 10 give? Which IN gives OUT 26?

6 Ben had some money. He spent $15 on a book. Then his dad gave him $10. Now he has $42. How much money did Ben have at first?

What your child learns

Two-step number machines and working backwards from the answer, with bar models.

Tip for you

Play "I double my number and add 3, and get 15". Swap roles and let your child make up the machine. Before class: have paper strips and counters ready, and write x cards (x 2, x 3, x 4, x 5, x 10) and + or - cards (1 to 20) on small pieces of paper.

Ask

"What is the opposite of each step?"

7 Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?

Go further

8 Ali, Ben and Siti share 42 stickers. Ben gets 2 more than Ali. Siti gets twice as many as Ali. How many stickers does each child get?

9 A machine does × 2, then − 4. Ravi puts a number in, then puts the OUT straight back in again. The final OUT is 12. What number did he put in first?

W3 LESSON 6 • MONEY Market Day

Start here

1 Write 275¢ in dollars and cents.

2 Pay exactly 65¢ with the fewest coins.

3 Pay exactly $17 with the fewest notes.

4 A pencil costs 45¢. Mei pays with a $1 coin. How much change does she get?

Think harder

5 Ben spent half of his money on a kite. Then he spent $4 on a ball. He had $6 left. How much money did Ben have at first? Tip: draw his money as a bar cut in half.

6 How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?

7 Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.

What your child learns

Paying with the fewest notes and coins, finding every way to pay, giving change and checking a budget.

Tip for you

Let your child pay at a hawker stall or shop and count the change before you check it. Before class: cut out the play coins and notes (kit pages) and keep the spare coins in the envelope.

Ask

"Is there a way with fewer coins?"

Go further

8 Ali has 10 coins. Every coin is a 10¢ coin or a 20¢ coin. They could be worth $1.50: how many of each coin is that? Write ALL the totals his 10 coins could be worth. Could they be worth $1.55? Explain.

9 Mei has $1. At the snack stall: chips 45¢, sweet 15¢, jelly 25¢, biscuit 30¢, lollipop 20¢. She wants as many DIFFERENT snacks as she can. Which should she buy?

W4 LESSON 7 • FRACTIONS
Fair Shares

Start here

1 A cake is cut into 5 equal pieces. Ben eats 2 pieces. What fraction of the cake does he eat?

2 How many quarters make 1 whole?

3 Which is bigger, 1/3 or 1/6?

4 3/10 + 4/10 = ?

Think harder

5 Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.

6 A ribbon is cut into 9 equal pieces. Mei uses 4 pieces and Siti uses 3 pieces. What fraction of the ribbon is left?

7 2/7 + ? = 1 whole

What your child learns

Halves, thirds and quarters as equal parts, comparing fractions, and adding parts of the same whole.

Tip for you

Share a roti prata or a pizza fairly. Ask if the pieces are really equal, and which is bigger: a third or a quarter. Before class: cut out the fraction strips (kit page) and have 4 paper strips of the same length.

Ask

"Are the parts the same size?"

Go further

8 Three children share a cake cut into 10 equal pieces. Each child gets a DIFFERENT number of pieces (at least 1), and the whole cake is shared out. Find ALL the ways to share it. Write each child's share as a fraction.

9 I am a fraction with 1 on top. I am smaller than 1/4 but bigger than 1/7. What could I be? Find all the answers.

W4 LESSON 8 • 2D AND 3D SHAPES
Shape Builders

Start here

1 How many quarter circles make a whole circle?

2 How many semicircles make a whole circle?

3 How many faces does a cube have?

4 How many squares of ALL sizes are in a grid 2 squares long and 2 squares high? (Small ones and one big one.)

Think harder

5 A stack of cubes has a bottom layer 4 cubes long and 3 cubes deep, a middle layer 3 cubes long and 2 cubes deep, and 1 cube on top. How many cubes are in the stack? Draw the layers.

6 How many squares are in a grid 3 squares long and 2 squares high?

What your child learns

Making and spotting shapes, semicircles and quarter circles, counting cubes in stacks, and nets.

Tip for you

Build a small tower with blocks or sugar cubes. Ask how many cubes there are, including the ones you cannot see. Before class: have scissors, 2 paper circles of the same size and about 12 small cubes ready, and cut out nets A to F (kit page).

Ask

"Did you count the hidden ones?"

7 Cut out nets E and F from the cut-out page and try folding them. Which one folds up into a cube? How do you know?

Go further

8 How many squares can you find in a grid 4 squares long and 3 squares high? Another grid, longer than it is high and at least 2 squares high, also has exactly 20 squares of all sizes. Find ALL the grids like that.

9 A square has both lines from corner to corner drawn. How many triangles can you find?

W5 Measure Masters

LESSON 9 • LENGTH, MASS, VOLUME AND TIME

Start here

1 One bag of flour balances 2 bags of sugar. A bag of sugar is 300 g. How heavy is the bag of flour?

2 A bucket holds 10 ℓ. It has 4 ℓ of water in it. Ali pours in 3 ℓ more. How many more litres are needed to fill it?

3 A ribbon is 90 cm long. Mei cuts off 25 cm. How long is the piece that is left?

4 A cartoon starts at 5:40 p.m. and lasts 30 minutes. What time does it end?

Think harder

5 Ben cuts a 20 m rope into 4 equal pieces. How long is each piece? How many cuts does he make?

6 Mei's family leaves home at 10:45 a.m. They reach Grandma's flat at 12:10 p.m. How long is the trip?

What your child learns

Balance scales, pouring puzzles with jugs, lengths end to end, and telling time and how long things take.

Tip for you

Use two cups of different sizes and a basin. Can your child measure exactly one cup of water a different way? Before class: make the coat-hanger balance (hanger, two paper cups, string), have a glue stick, a spoon and 20 coins, two different-size cups with a jug of dry rice or sand (tape marks at 3 and 5), and cut out the paper clock.

Ask

"What do you know for sure from the scale?"

7 3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?

Go further

8 A basket with 4 apples is 700 g. The same basket with 2 apples is 500 g. How heavy is the empty basket?

9 A rope 40 cm long is cut into pieces 10 cm long. Each cut takes 2 minutes. How long does the cutting take?

W5 Challenge Day

LESSON 10 • MIXED PUZZLES

Start here

1 Use the digit cards 5, 1 and 9 once each. Make the greatest 3-digit number and the smallest 3-digit number.

2 In a picture graph, each picture stands for 2 children. Apples has 4 pictures. How many children chose apples?

3 Ali has 3 bags with 5 marbles in each bag. Mei has 4 bags with 4 marbles in each bag. Who has more marbles? How many more?

4 Make 80¢ with exactly 3 coins. You can use 5¢, 10¢, 20¢ and 50¢ coins.

Think harder

5 30 children chose a fruit. In the picture graph each picture stands for 5 children. Apple has 2 pictures and banana has 3 pictures. The mango row is missing. How many pictures should the mango row have?

6 How many 3-digit numbers have digits that add up to 3? List them all.

What your child learns

Choosing how to start a puzzle: draw it, list it or make a table, plus picture graphs and logic grids.

Tip for you

When your child is stuck, ask "Can you draw it?" before giving any other hint. Before class: have the digit cards, play coins, paper strips, a pencil and a small whiteboard ready.

Ask

"How did you start? What would you try next time?"

7 A bus had some people on it. At the first stop half of the people got off. At the second stop 6 people got on. Now there are 14 people on the bus. How many people were on the bus at the start? Draw a bar to start.

Go further

8 Use three DIFFERENT digit cards to make 3-digit numbers. With the cards 0, 3 and 8 you can make exactly 4 numbers. With the cards 3, 5 and 8 you can make 6. Which sets of three different cards make exactly 4 numbers? Test some sets and write a rule. Does your rule always work?

9 Ali, Mei, Siti and Ben each go to school a different way: bus, MRT, car or walking. Clues: Siti walks. Ben does not go by bus or by car. Mei does not go by car. Who goes by what? Then take away ONE clue at a time. Which clue could you take away and still have exactly one answer? How many answers does each clue-less puzzle have?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Digit Detectives

1. 538
2. 10 (1 ten)
3. 557
4. 593
5. 6 numbers: 158, 185, 518, 581, 815, 851.
6. 3 numbers: 447, 474, 744. The two 4s look the same, so swapping them does not make a new number.
7. 446 (14 tens = 1 hundred and 4 tens).
8. 24. Leave out one card (4 choices); each set of 3 cards makes 6 numbers: 6 + 6 + 6 + 6 = 24.
9. 15. Hundreds digit 1: 5 numbers; 2: 4; 3: 3; 4: 2; 5: 1. 5 + 4 + 3 + 2 + 1 = 15.

Week 1 • Lesson 2: Missing Digit Mysteries

1. 6 (365 + 214 = 579)
2. 3 (638 – 205 = 433)
3. 350
4. 135
5. Corners 140, 150, 160. Between 140 and 150: 130. Between 150 and 160: 110. Between 160 and 140: 120 (each side: 140 + 130 + 150 = 420, 150 + 110 + 160 = 420, 160 + 120 + 140 = 420).
6. 557 + 348 = 905 (ones 7 + 8 = 15; tens 5 + 4 + 1 = 10; hundreds 5 + 3 + 1 = 9).
7. 721 – 458 = 263. Check: 263 + 458 = 721.
8. 531: 654 – 123.
9. 200 (200 + 150 = 350 and 550 – 200 = 350).

Week 2 • Lesson 3: Legs, Wheels and Arrays

1. 10 wheels (2, 4, 6, 8, 10).
2. 20 chairs (4 × 5 = 20).
3. 6 sweets (18 ÷ 3 = 6).
4. 14 legs (7 × 2 = 14).
5. 3 ways: 7 bicycles and 1 tricycle (14 + 3); 4 bicycles and 3 tricycles (8 + 9); 1 bicycle and 5 tricycles (2 + 15). 0, 2 or 4 tricycles leave an odd number of wheels, which bicycles cannot make, and 6 or more tricycles need 18 wheels or more.
6. 6 hens and 3 goats. Start with 9 hens (18 legs); each swap to a goat adds 2 legs; 3 swaps make 24. Check: 12 + 12 = 24.
7. 8 each (4 × 8 = 32) and 2 left over.
8. No. 5, 7, 11, 13, 17, 19, 23 and 29 cannot (8 numbers): only 1 row works.
9. 3 ways (bikes, trikes, cars): 3, 2, 4; 2, 4, 3; 1, 6, 2.

Week 2 • Lesson 4: Pattern Machines

1. 16, 20, 24
2. Blue (every even place is blue).
3. 30 fingers (5, 10, 15, 20, 25, 30).
4. A circle. 9 ends a group of 3, and each group ends with a circle.
5. 12 circles (30 shapes is 6 groups of 5, with 2 circles in each group).
6. A. 24 ends a group of 4 (4 × 6 = 24), so 25 starts again with A.
7. 12 yellow beads (18 beads is 6 groups of 3, with 2 yellow in each).
8. At 15 seconds and at 30 seconds (numbers in both the 3 and 5 times tables).
9. 9 houses use 37 sticks, with 3 left over (10 houses would need 41).

Week 3 • Lesson 5: Mystery Number Machines

1. 9
2. 16
3. 55
4. 6 (20 – 2 = 18, 18 ÷ 3 = 6)
5. × 3, then + 5. IN 10 gives OUT 35. OUT 26 comes from IN 7 (26 – 5 = 21, 21 ÷ 3 = 7).
6. $47 (42 – 10 = 32, 32 + 15 = 47)
7. 36 (30 – 12 = 18 is half, 18 + 18 = 36)
8. Ali 10, Ben 12, Siti 20 (40 ÷ 4 = 10)
9. 6 (back twice: 12 → 8 → 6)

Week 3 • Lesson 6: Market Day

1. $2.75
2. 50¢ + 10¢ + 5¢ (3 coins)
3. $10 + $5 + $2 (3 notes)
4. 55¢
5. $20 ($6 + $4 = $10 is the half he kept after the kite, so the other half is also $10: $10 + $10 = $20).
6. 10 ways (1 + 3 + 6)
7. 2 sets: $1 + 10¢ + 10¢ + 5¢, and 50¢ + 50¢ + 20¢ + 5¢
8. Five and five. Totals $1.00 to $2.00, 10¢ apart (11). Not $1.55.
9. Sweet, lollipop, jelly and biscuit (90¢). Five snacks cost $1.35.

Week 4 • Lesson 7: Fair Shares

1. 2/5
2. 4
3. 1/3 (3 parts are bigger than 6 parts)
4. 7/10
5. 1/12, 1/5, 1/3, 1/2
6. 2/9 (4 + 3 = 7 pieces used, 9 – 7 = 2 left)
7. 5/7
8. 4 ways (tenths): 1, 2, 7; 1, 3, 6; 1, 4, 5; 2, 3, 5.
9. 1/5 and 1/6

Week 4 • Lesson 8: Shape Builders

1. 4
2. 2
3. 6
4. 5 (4 small + 1 big)
5. 19 (12 + 6 + 1)
6. 8 (6 small + 2 made of 4 small ones)
7. E folds into a cube (it is a staircase of squares, one of the 11 cube nets). F does not: the row of 5 squares wraps round and the 1st and 5th squares land on the same face.
8. 20 (12 + 6 + 2). Also a 7 by 2 grid.
9. 8 (4 small + 4 made of 2 small ones)

Week 5 • Lesson 9: Measure Masters

1. 600 g (300 + 300)
2. 3 ℓ (4 + 3 = 7, and 10 – 7 = 3)
3. 65 cm
4. 6:10 p.m.
5. 5 m each (5 + 5 + 5 + 5 = 20). 3 cuts: the last piece needs no cut.
6. 1 hour 25 minutes (10:45 to 11:00 is 15 minutes, 11:00 to 12:00 is 1 hour, 12:00 to 12:10 is 10 minutes).
7. Apple 100 g, pomelo 300 g (5 apples balance 500 g).
8. 300 g (the 2 extra apples are 200 g)
9. 6 minutes (4 pieces need only 3 cuts)

Week 5 • Lesson 10: Challenge Day

1. 951 and 159
2. 8 children
3. Mei, by 1 (Ali 15, Mei 16)
4. 50¢ + 20¢ + 10¢ (the only way)
5. 1 picture (30 – 10 – 15 = 5 children)
6. 6: 102, 111, 120, 201, 210, 300
7. 16 (14 – 6 = 8 is half, 8 + 8 = 16).
8. Exactly the sets with a 0 card: 0, a, b make a0b, ab0, b0a, ba0. Sets without 0 make 6.
9. Ali car, Mei bus, Siti walks, Ben MRT. No clue can go (8, 4 or 2 answers).